Evaluating the Impact of Feature Selection on Ensemble Models for Multiclass Heart Disease Prediction

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***Abstract*—This study discusses the impact of different feature selection techniques on the performance of ensemble learning algorithms, namely Random Forest and XGBoost, in multiclass heart disease prediction. A real-world clinical dataset with class imbalances was used, which were resolved using the Synthetic Minority Oversampling Technique (SMOTE). Three different feature selection techniques—Chi-Square, Mutual Information, and SelectFromModel—were tested against baseline models without feature selection. The models were trained with hyperparameter tuning, and their performance metrics were compared using accuracy, precision, recall, and F1-score. The results showed that both Random Forest and XGBoost achieved optimal performance without feature selection, with the Random Forest model achieving the highest accuracy of 57.1%. SHAP interpretations showed that erythrocyte, hemoglobin, and hematocrit were the major features that determined model predictions. These results suggest that tree-based ensemble techniques may have an inherent ability for feature relevance handling, reducing the need for explicit selection methods. This study adds to the knowledge and efficiency of predictive modeling in clinical diagnostics.**

***Index Terms*—Ensemble Learning, Random Forest, XGBoost, Feature Selection, Heart Disease Prediction, SHAP**

I. Introduction

Cardiovascular illness is a leading cause of worldwide mortality, reported by the World Health Organization (WHO) to account for nearly 17 million deaths annually. The accurate and timely detection of these disorders is crucial to reducing mortality rates, but this evaluation often depends on the knowledge and skill of the medical practitioner, leading to potential misdiagnoses and increased medical costs [1].

In recent times, Machine Learning (ML) techniques have been increasingly regarded as very potent predictive model tools, especially when applied to the diagnosis of cardiovascular diseases [2]. Such increased emphasis on Machine Learning is partially due to the remarkable ability of Machine Learning to handle large and complicated datasets, uncover hidden patterns, and obtain results often beyond the reach of human analysis [3].

Of the wide variety of machine learning algorithms, ensemble learning is often considered one of the most powerful methods in the field. Ensemble learning combines several base learners to create a combined and stronger model. As such, ensemble learning offers many advantages that enhance its effectiveness as a method for enhancing predictive power on a variety of tasks [4].

Among the most extensively used ensemble algorithms are Random Forest (RF) and Extreme Gradient Boosting (XGBoost). Random Forest uses a bagging method and builds many individual decision trees, and each of these contributes to a final classification output [5]. In contrast, XGBoost uses a boosting approach and builds decision trees sequentially, with each tree designed to minimize the errors of the previous ones. This iteration helps to gradually minimize the predictive error over many learning iterations [6].

Despite the empirically proven effectiveness of ensemble algorithms like Random Forest and Extreme Gradient Boosting (XGBoost), their effectiveness is largely dependent on the features and nature of the input data. Clinical datasets, especially those related to cardiac diseases, tend to have high-dimensional spaces with a very high percentage of irrelevant or redundant features, and are prone to imbalance in general. These issues can lead model overfitting and biased predictions [7]. To overcome these challenges, methods like feature selection are utilized to maintain discriminating features, while data sampling methods like the Synthetic Minority Over-sampling Technique (SMOTE) are used to correct class imbalance and improve model robustness [8].

This study investigates the influence of three different feature selection methods—Chi-Square, Mutual Information, and SelectFromModel—on the performance of Random Forest and XGBoost classifiers in terms of multiclass heart disease classification. The Synthetic Minority Over-sampling Technique (SMOTE) is used to counteract the imbalance in the dataset, and hyperparameter tuning improves prediction accuracy. In addition, SHAP (SHapley Additive Explanations) is used to explain how features contribute to the models' predictions.

Against the limitations of the clinical data available, this study focuses on the hematologic features remaining with clin-

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ical significance. The following sections of this manuscript are organized in this way: the following section are an overview of previous research pertinent to cardiovascular disease. Section III outlines the methods used in this study. Section IV interprets the results and the models used. Section V summarizes the conclusions from the findings of the research.

II. Related Works

The field of Machine Learning (ML) has received considerable interest in the medical field, particularly in the prediction and diagnosis of cardiovascular disease. Various research studies have indicated that ML models have the potential to identify patterns in clinical data that often escape human analysis. For example, Snigdha Datta et al. utilized Logistic Regression to predict cardiovascular disease with an accuracy of 81% [9].

Among the many algorithms studied, ensemble-based algorithms like Random Forest and XGBoost have always proved to be highly effective in cardiovascular disease prediction. Madhumita Pal and Smita Parija reported that the Random Forest model achieved an accuracy of more than 86.9% in predicting heart disease [10]. Similarly, Juan Carlos Yang found that XGBoost, when carefully tuned with appropriate hyperparameters, outperformed conventional models by achieving more than 88% accuracy and even reaching up to 91% accuracy in certain experiments, thus proving its credibility in solving medical datasets [11].

To further improve the precision of predictions, researchers have used feature selection methods with the aim of reducing dimensionality and eliminating unnecessary features. According to Bilal Ahmad et al., Mutual Information (MI) and Chi-Square considerably increase the performance of classifiers while maintaining computational efficiency [12]. Similarly, Saikiran Kutikuppala et al. utilized the SelectFromModel feature selection method to simplify the process of classification and maximize efficiency through the systematic selection of the most relevant features [13].

One common problem faced in medical datasets is the occurrence of class imbalance, which causes overfitting of the model and biased prediction generation [7]. To overcome this issue, numerous oversampling methods have been extensively used, of which the most recognized one is the Synthetic Minority Over-Sampling Technique (SMOTE). Anitha Juliette Albert et al. proved that the use of SMOTE resulted in significant improvements in terms of accuracy, precision, recall, and F1 score compared with non-sampling datasets [14].

There has been increased emphasis on interpretability in the field of medical machine learning in recent years. SHAP (SHapley Additive Explanations) is a remarkable tool that provides clear and theory-driven evaluations of feature importance. According to Amira et al., SHAP ensures that the contribution of every feature is fairly allocated, thus providing accurate and interpretable explanations for model predictions. Its ability to provide both global and local interpretability makes it particularly beneficial for unraveling complex machine learning models, especially in high-stakes settings like medical diagnosis [15].

While many studies have applied ensemble models or feature selection separately, few have systematically examined their combined impact on multiclass heart disease classification. Most prior work has focused on binary classification, leaving multiclass prediction underexplored. This study addresses these gaps by assessing how Chi-Square, Mutual Information, and SelectFromModel affect the performance of Random Forest and XGBoost, supported by SMOTE and hyperparameter tuning.

III. METHODOLOGY

The main aim of this study is to evaluate the impacts of three Feature selection techniques—Chi-Square, Mutual Information, and SelectFromModels to test the performance of Random Forest and XGBoost for multi-class prediction of heart disease. The overall framework shown in Figure 1 includes data preprocessing, SMOTE for handling class imbalance, feature selection, model building with hyperparameter optimization, performance evaluation, and SHAP interpretation. Each step is designed to Evaluate how feature selection affects the accuracy of the model and determine the essential clinical factors leading to cardiovascular disease.

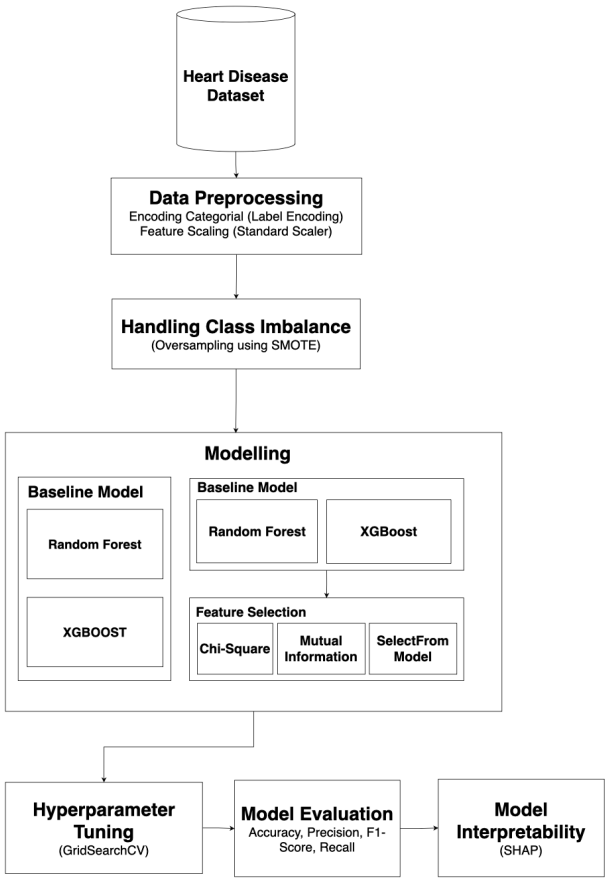


Fig. 1. Research proposed framework

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A. Dataset Description

The data used in the current study come from a medical clinic located in West Jakarta and can be accessed at https://zenodo.org/records/15172095. The dataset in question is obtained from a standalone cardiovascular hospital with expertise in the diagnosis and treatment of heart diseases. The autonomy ensures that a large percentage of patients who report to the hospital have confirmed or suspected cardiac conditions, thus making the dataset especially representative and relevant to research in the prediction of heart disease. The dataset contains ten different attributes and 4,752 instances, including the class label. The label attribute, which is represented as "icd pr," indicates the precise type of angina pectoris the patient has been diagnosed with. The World Health Organization (WHO) classifies unstable angina based on the International Classification of Diseases, 10th Revision (ICD-10), and assigns it I20.0. This class includes crescendo angina, de novo effort angina, and provoked effort angina. In addition, the code I20.8 covers other types of angina pectoris, which include effort angina and stenocardia, while I20.9 is reserved for unspecified angina pectoris, including general angina (NOS - Not Otherwise Specified) [16]. The attributes used in this study are described as follows:

- **Gender**: Categorical variable indicating patient sex (male or female).
- **Age**: The patients' ages range from 20 to 94 years.
- **Erythrocyte**: Red blood cells responsible for oxygen transport. The erythrocyte sedimentation rate (ESR) can serve as a biomarker for inflammatory responses associated with cardiovascular conditions [17], [18].
- **Hematocrit**: Hematocrit levels are indicative of the proportion of red blood cells in blood volume and can be used to assess conditions such as anemia or polycythemia. Higher hematocrit levels have been associated with increased risk of coronary heart disease (CHD) [19], [20].
- **Hemoglobin**: A protein in red blood cells that carries oxygen. Low hemoglobin levels (e.g., < 14g/dL for STEMI or < 11g/dL for NSTE-ACS) are associated with increased cardiovascular risk [21].
- **Leukocytes**: White blood cell count. An elevated count can indicate inflammation and higher cardiovascular risk due to neutrophil and monocyte activation [22]. Leukocytes are also biomarkers for infections, inflammation, allergies, and leukemia [23].
- **Thrombocyte**: A high level of thrombocytes in the patient's blood indicates an increased risk of coronary heart disease [24]. Any increase or decrease in the thrombocyte level from the normal range, along with any dysfunction, is expected to lead to complications; thus, the determination of the level of thrombocytes can help in the diagnosis of certain medical conditions [25].
- **Hermch (MCH)**: Mean corpuscular hemoglobin, indicating the average amount of hemoglobin per red blood cell [26].
- **Khermchc (MCHC)**: Mean corpuscular hemoglobin concentration (MCHC) measures the concentration of hemoglobin in red blood cells [26], [27].

B. Data Preprocessing

The preprocessing phase included the conversion of categorical variables and normalization of numerical data. The 'gender' feature was expressed in binary form, assigning 1 to females and 0 to males. Target variable 'icd_pr' was transformed using LabelEncoder for multiclass classification. In addition, features were standardized using the application of StandardScaler to normalize their scales and improve learning stability. These operations made the dataset comply with machine learning algorithm requirements while, at the same time, preventing the possibility of biased learning due to scale differences among features.

C. Handling Class Imbalance

One challenge in this study was class imbalance in the target feature that can cause overfitting and bias toward the majority classes, thus reducing performance on minority classes [7]. In the dataset, 44.9% of instances belong to class I20.0, 31.4% to I20.8, with only 23.7% of it being due to I20.9, highlighting the need for corrective measures.

To remedy the problem under study, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the dataset. SMOTE operates by generating synthetic instances for the minority classes through interpolation between current samples in the minority class and their k-nearest neighbors [28]. This worked to effectively balance the distribution across the classes, with each class being approximately 33.3% of the population in the sample.

To formalize the SMOTE approach, synthetic samples are obtained using the following interpolation equation:

$$x_{\text{new}} = x_i + \lambda \cdot (x_{zi} - x_i), \quad \lambda \sim \mathcal{U}(0, 1)$$

where x_i is an observation from the minority class, and x_{zi} is one of its k-nearest neighbors algorithm, where λ is an arbitrarily drawn random variable. An uniform distribution between 0 and 1. This description was initially presented as a method, called SMOTE, by Chawla et al. [35]

D. Feature Selection Techniques

The current study applied three different feature selection methods—Chi-Square, Mutual Information, and SelectFromModel—to analyze their performance in selecting the most relevant features for classifying angina pectoris into different types. Each of the main types of feature selection was catered by a particular method: Chi-Square was used as a standard filtering method, Mutual Information was an entropy-based filter that is good at identifying both linear and non-linear associations, and SelectFromModel was an embedded method that makes use of internal model scores to determine significance. This systematic approach allows for a comparison between classic univariate statistical methods and model-based approaches that are better at handling complex interactions.

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First, the input features were normalized using MinMaxScaler to enable the use of the Chi-Square approach, since the method requires non-negative input variables. The five most important features were determined based on the highest Chi-Square values. For the Mutual Information approach, normalized data was used to determine the mutual dependence of each feature and the target variable, again selecting the top five features. Under the SelectFromModel approach, the pre-trained Random Forest and XGBoost model was used to determine feature importance scores, making it easy to select features with importance values above the median.

E. Modeling

Random Forest and XGBoost are widely recognized as robust models for clinical predictions [31], [32]. As methods in ensemble learning, they naturally encompass feature selection processes that effectively counteract problems regarding multicollinearity, noise, and irrelevant variables without requiring extra preprocessing. The non-parametric nature of these models allows detecting non-linear relationships, making them more robust against the outliers commonly seen in clinical data [33]. When properly tuned, these models have lower probabilities of overfitting, even for scenarios involving class imbalances or small sample sizes [34], making them especially well-suited to heterogeneous and noisy data.

In the current study, Random Forest and Extreme Gradient Boosting (XGBoost) were selected because of the proven success of these techniques on complex classification problems and the ability to scale to high-dimensional datasets. Random Forest follows a bagging approach, building multiple decision trees whose outputs are averaged to yield the final prediction [5], while XGBoost has a boosting approach, gradually adding trees to correct the faults of previous trees and thus systematically reducing prediction error [6].

All models were trained on the SMOTE-balanced dataset using all feature selection variants, namely Chi-Square, Mutual Information, and SelectFromModel, and a baseline without feature selection. This enabled a direct comparison of their individual impacts on performance. Hyperparameters were optimised using grid search to ensure the optimal model settings.

For enhancing the models' interpretability, SHAP (SHapley Additive Explanations) was applied on two different setups: the baseline Random Forest model without feature selection and the XGBoost model with Chi-Square feature selection. The Random Forest model, which includes all the available features, provides a holistic view of the contribution of each clinical attribute to the predictions made. On the other hand, the XGBoost model, which uses feature selection, provides a more focused and understandable explanation that depends only on the variables found to be statistically significant. This two-pronged assessment allows for a meaningful comparison between full and reduced feature sets and, in doing so, highlights the attributes that are key predictors of different types of angina pectoris.

F. Hyperparameter Tuning

In order to maximize the performance and enable a fair comparison between the models, hyperparameter tuning was applied on the Random Forest and XGBoost classifiers using GridSearchCV, a module found within the Scikit-learn library. The GridSearchCV approach is meant to find the best parameters for a classifier by tuning specific hyperparameters which are not learned automatically when training the model, thus improving the ability of the model to make accurate predictions on new, never-before-seen data [29].

For the Random Forest model, the optimized hyperparameters were n_estimators = [50, 100, 150] and max_depth = [None, 5, 10]. For the XGBoost model, the search space for the parameters included n_estimators = [50, 100, 150] and max_depth = [3, 5, 10]. Both models used a grid search approach with 3-fold cross-validation, with accuracy as the main evaluation criterion. For XGBoost, the use_label_encoder=False parameter setting was used to avoid the use of outdated features, and the 'mlogloss' evaluation metric was used to better support multiclass classification problems. The best parameters found for each model were then applied uniformly across all experimental trials.

G. Evaluation Metrics

The model's performance was evaluated utilizing a range of measures. Accuracy was the overall proportion of instances correctly predicted, while precision, recall, and F1-score were calculated for each class separately to provide a complete assessment—a critical factor within medical datasets, where misclassification can have severe consequences. A confusion matrix was also generated to illustrate correct and incorrect predictions in relation to the three classes of angina pectoris. All the analysis steps were performed using functions from scikit-learn.

IV. Results and Discussions

A. Performance Evaluation of Models

The performance of both the XGBoost and Random Forest classifiers was compared over a variety configurations, including baseline (no feature selection) and three feature selection methods: Chi-Square, Mutual Information, SelectFromModel. Evaluation metrics included accuracy, precision, recall, and F1-Score.

Table I shows that Random Forest performed better than XGBoost in terms of overall accuracy for all tests. The baseline Random Forest model with no feature selection produced a peak accuracy of 57.1%, while the same configuration for XGBoost produced an accuracy of 56.4%.

For this analysis, strong emphasis was placed on macro-averaged measures such as precision, recall, and F1-score to balance off against somewhat imbalanced class distributions. This framework ensures that minority class performances like I20.9 are still focal and not overshadowed by majority class prevalence.

It is significant to note that the models without including feature selection performed better. This is an indication that

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inherent methodology used by ensemble methods to handle feature importance surpasses that provided by explicit feature selection methods. This finding is congruent with previous work published by Yang et al., which showed that ensemble models based on trees, like Random Forest and XGBoost, perform feature selection automatically during tree construction and reduce the need for explicit preprocessing mechanisms [30].

TABLE I. Performance Comparison of Ensemble Models with and without Feature Selection

No.	Model	Precision	Recall	F1-Score	Accuracy (%)
1	Random Forest	0.57	0.57	0.57	57.1
2	XGBoost	0.56	0.56	0.56	56.4
3	Random Forest + Chi-Square	0.56	0.57	0.56	56.5
4	Random Forest + Mutual Info	0.56	0.56	0.56	56.2
5	Random Forest + SelectFromModel	0.55	0.56	0.55	55.7
6	XGBoost + Chi-Square	0.54	0.54	0.54	54.2
7	XGBoost + Mutual Info	0.53	0.53	0.53	53.3
8	XGBoost + SelectFromModel	0.55	0.55	0.55	54.8

A statistical test was conducted to determine whether the differences in performance were statistically significant. The one-way ANOVA of the XGBoost results revealed no significant differences between different configurations, as indicated by an F-statistic of 0.3643 and a P-value of 0.7792. Nevertheless, as shown in Table II, the 95% confidence interval for the model that did not use feature selection was between 0.5354 and 0.6486, which did not overlap with the others, implying a possible performance gain. This claim is further supported by the outcome of the paired t-test shown in Table III, which shows that all the comparisons conducted between the non-feature selection model and the three feature selection methods had p-values less than the 0.05 significance level, hence implying statistically significant differences from what was observed.

TABLE II. XGBoost Model Accuracy and 95% Confidence Intervals by Feature Selection Method

Model	Mean Accuracy	95% CI (Lower)	95% CI (Upper)
No Feature Selection (No FS)	0.5920	0.5354	0.6486
Chi-Square	0.5619	0.4949	0.6289
Mutual Information	0.5583	0.4894	0.6272
SelectFromModel	0.5556	0.4954	0.6159

TABLE III. Statistical Comparison of XGBoost Performance With and Without Feature Selection Using Paired t-Test

Comparison	t-statistic	p-value
No Feature Selection vs Chi-Square	3.3360	0.0087
No Feature Selection vs Mutual Information	3.4223	0.0076
No Feature Selection vs SelectFromModel	6.6423	0.0001

On the other hand, the Random Forest approach manifested a more stable performance irrespective of the variable settings utilized. ANOVA test results provided an F-statistic value of 0.1762 and a P-value of 0.9118, hence indicating a lack of statistically significant differences in the performance between the different feature selection algorithms. As shown in Table IV, the 95% confidence intervals for each setup bear significant overlap. However, the setup without feature selection shows marginally higher values at both the lower and upper ends of the interval. The results of the paired t-test, as shown in Table V, also support this claim, as all the comparisons yielded p-values larger than 0.05, hence pointing to the statistical insignificance of the observed differences in performance.

TABLE IV. Random Forest Model Accuracy and 95% Confidence Intervals by Feature Selection Method

Model	Mean Accuracy	95% CI (Lower)	95% CI (Upper)
No Feature Selection	0.6048	0.5470	0.6626
Chi-Square	0.5814	0.5137	0.6491
Mutual Information	0.5814	0.5155	0.6473
SelectFromModel	0.5969	0.5369	0.6569

TABLE V. Statistical Comparison of Random Forest Performance With and Without Feature Selection Using Paired t-Test

Comparison	t-statistic	p-value
No Feature Selection vs Chi-Square	2.1445	0.0606
No Feature Selection vs Mutual Information	2.1621	0.0589
No Feature Selection vs SelectFromModel	0.9460	0.3689

These results indicate that whether feature selection works well or not is very much dependent on the learning algorithm applied. It is also advised to apply bigger samples or other procedures such as bootstrap confidence intervals in order for the findings to be robust.

B. Effect of SMOTE on Model Performance

To counter the class imbalance detected in the dataset, Synthetic Minority Oversampling Technique (SMOTE) was applied before the phase where the model was trained. Table VI contains the summary comparison between Random Forest, XGBoost, both with and without applying SMOTE, while omitting the consideration of feature selection.

Results clearly show that use of the SMOTE method had resulted in considerable improvement in all the performance parameters of both the models. For Random Forest, accuracy improved from 45.5% to 57.1%, while the F1-score improved from 0.44 to 0.57. Similarly, XGBoost improved its accuracy from 48.2% to 56.4%, with corresponding improvement in the F1-score from 0.44 to 0.56.

The improvements were especially evident in the recall measures, which show that SMOTE strongly increases the ability of predictive models to effectively detect instances in minority classes. As a result, although predictive accuracy is boosted, the fairness and robustness of the classification system are also increased, making it extremely suitable for use in clinical practice with imbalanced medical data.

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TABLE VI. Effect of SMOTE on the Performance of Random Forest and XGBoost Without Feature Selection

No.	Model	Precision	Recall	F1-Score	Accuracy (%)
1	Random Forest (No SMOTE)	0.45	0.45	0.44	45.5
2	Random Forest (With SMOTE)	0.57	0.57	0.57	57.1
3	XGBoost (No SMOTE)	0.48	0.44	0.44	48.2
4	XGBoost (With SMOTE)	0.56	0.56	0.56	56.4

C. Confusion Matrix Analysis

To evaluate the predictive performance for every class, confusion matrices were created for four different settings, as shown in Figure 2: Random Forest without feature selection, XGBoost without feature selection, XGBoost with Chi-Square, and XGBoost with Mutual Information. These settings include both baseline options as well as those where feature selection was applied to the two ensemble methods.

The Random Forest baseline produced relatively balanced predictions across the three angina classes but with a high misclassification rate. XGBoost without feature selection slightly improved identification of classes I20.8 and I20.9 but had high false positives for class I20.0. Applying Chi-Square or Mutual Information reduced true positive rates and increased misclassifications in most classes, confirming that external feature selection reduced performance compared to using the full feature set.

Overall, the confusion matrices visually reinforce the quantitative findings above by illustrating class-level prediction patterns.

D. Model Interpretability using SHAP

SHAP (SHapley Additive Explanations) refers to a method that defines a class of additive importance functions for features to produce an individual solution that satisfies a number of basic properties, such as local accuracy, dealing with missing values, and consistency. The framework is based on Shapley values, which are rooted in cooperative game theory, and seeks to allocate fairly the impact of each feature on the difference between the true prediction made by the model and the average prediction over all possible combinations of features. The method was introduced by Lundberg and Lee [36]. Computation of the Shapley value of a feature i is performed using the following expression:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \left[f(S \cup \{i\}) - f(S) \right]$$

F is the set of all input features. S is any subset of F that does not include feature i. $f(S)$ is the model-predicted value when the model utilizes only the features in subset S.

To improve the interpretability of the models and to explore the impact of individual features on predictions, SHAP (SHapley Additive Explanations) was applied to two different model configurations: a Random Forest model that did not use feature selection, and an XGBoost model that used Chi-Square feature selection. The aim was to identify and compare

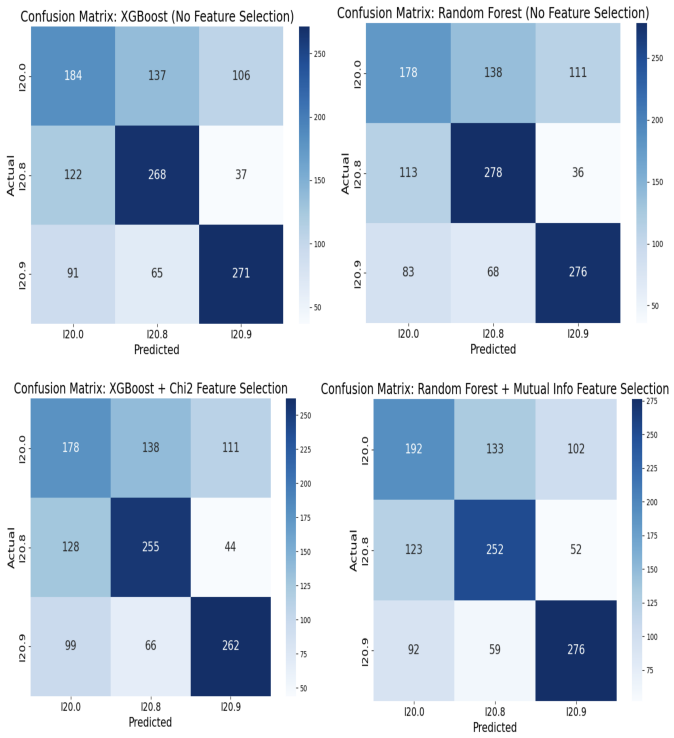


Fig. 2. Confusion matrices of selected models: (a) Random Forest (No FS), (b) XGBoost (No FS), (c) XGBoost + Chi2, (d) XGBoost + Mutual Information

the most significant features in each model and determine if the Random Forest's internal feature selection mechanism is different from the externally applied features used in XGBoost.

As shown in Figure 3, both models rank erythrocytes, hemoglobin, and hematocrit as the top features relying on SHAP values. However, there is a clear difference in their distributions: XGBoost has high overall SHAP effects, reflecting a higher reliance on a few dominant features, while Random Forest shows a more balanced contribution from its dominant features, as in the nature of feature bagging. This suggests that while both models give prominence to similar biological markers, their interpretive models and sensitivity to these markers are different, which can have implications for clinical decision-making and the reliability of the models.

E. Discussion and Implication

This study investigated the effect of different feature selection methods on ensemble learning models for predicting multiclass heart disease, specifically comparing configurations of Random Forest and XGBoost both with and without feature selection. The top-performing models did not use formal feature selection; in particular, the standard Random Forest yielded the highest accuracy of 57.1%, which outperformed all variations that used Chi-Square, Mutual Information, or SelectFromModel feature selection. The same trend was noted with XGBoost. These results suggest that in low-dimensional settings, the benefits derived by feature selection are minimal;

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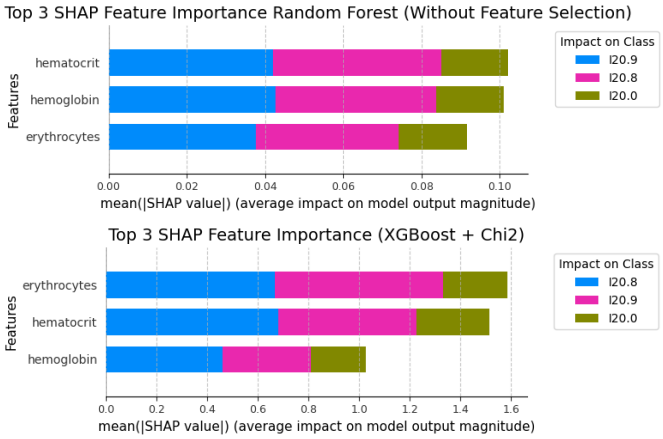


Fig. 3. Top 3 SHAP Feature Importance: (Top) Random Forest without feature selection; (Bottom) XGBoost with Chi-Square selection.

in this study, the main aim was to determine whether different selection methods would influence results rather than optimizing computational efficiency.

The findings indicate that tree-based ensembles like Random Forest and XGBoost inherently identify and weight important features during training, reducing the need for external preprocessing, which may inadvertently remove valuable variables. This aligns with Yang et al., who reported that such models perform built-in feature selection during tree generation, minimizing reliance on explicit preprocessing frameworks [30].

Moreover, the SHAP analysis showed that erythrocyte counts, hemoglobin concentrations, and hematocrit measurements were always significant in model predictions. These clinically relevant variables, linked with known cardiovascular indexes, serve to further support correspondence between model and existing knowledge in the domain. Importantly, even without explicit feature selection toward known targets, ensemble methods automatically offset these predictors. This property reduces the need for complex preprocessing, whereas SHAP-based interpretations provide a means for relating outputs to clinical expertise, enhancing explainability and building confidence in medical decision support.

The model's accuracy, measured at a relatively modest 57.1%, is thus limited by the dataset's sole emphasis on hematological markers like red blood cell counts, hemoglobin levels, and hematocrit values. Though these markers are of clinical interest in certain limited circumstances, they represent only a small subset of the many variables that may impact cardiovascular disease. Notably, the analysis did not comprise other key risk factors, such as blood pressure measurements, cholesterol levels, electrocardiogram findings, and disease familial history, thus limiting the model's ability to realize high predictive capacity.

Despite this fact, the level of accuracy achieved should not be ignored completely. It illustrates the possibility of using basic and easily available blood test information for the initial assessment of cardiovascular disease in resources-poor settings. This study emphasizes the capability of ensemble learning methods in deriving diagnostic information from limited datasets, which will be a useful and easily applicable tool for making risk assessments. The value of this study lies less in the diagnostic accuracy of the predictors and more in the development of a clinically meaningful, interpretable, and practical machine learning tool within the context of cardiovascular disease.

V. Conclusion

The major aim of this study was to evaluate the impact of different feature selection methods on the performance of ensemble machine learning models namely, Random Forest and XGBoost in the prediction of multiclass diagnoses of heart disease. Experimental design used a real clinical dataset with an uneven class distribution that was counteracted by the use of SMOTE. Various feature selection approaches, such as Chi-Square, Mutual Information, and SelectFromModel, were tested and compared with baseline models without the use of feature selection.

Results demonstrated both algorithms were optimized without feature selection, indicating both algorithms' built-in feature evaluation functions were effective for identifying relevant features. Without feature selection, Random Forest obtained the optimal accuracy of 57.1%, while XGBoost obtained an accuracy of 56.4%. By comparison, those models with feature selection showed somewhat lower performance. By SHAP analysis, erythrocyte count, hemoglobin level, and hematocrit were identified as the key predictors for both algorithms, paralleling already established clinical predictors for cardiac disease.

The findings highlight the importance of evaluating whether there is a need for external feature selection when applying tree-based ensemble methods. Additionally, the use of interpretable models provides essential insights that enable medical decision-making, thus increasing model transparency and building confidence in the clinical environment. The study is, however, limited by the inherent characteristics of the dataset since it was obtained from a single source and included a limited range of diagnostic groups.

Future research activities may involve investigating the integration of a wider variety of clinical features, utilize datasets acquired from diverse healthcare organizations, and investigate the use of deep learning models combined with explainability techniques. These improvements can enhance diagnostic accuracy and the effectiveness of decision support systems in the healthcare industry.

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